datetime Module

```
a module to handle data and time
```

Getting datetime Information

```
from datetime import datetime
now = datetime.now()
now.day
now.month
now.year
now.hour
now.minute
now.second
now.timestamp() # number of seconds elapsed from 01/01/1970 UTC
datetime(2020, 1, 1) #2020-01-01 00:00:00
```

strftime Formatting Date Output

```
from datetime import datetime
now = datetime.now()
time = now.strftime("%m/%d/%Y, %H:%M:%S")
#string for time
#mm/dd/YY H:M:S
```

> **Important**
>
> more info in documentation

strptime String to Time

```
date_string = "5 December, 2019"
date_object = datetime.strptime(date_string,  "%d %B. %Y")
print(date_object) #2019-12-05
```

date From datetime

```
limited to calendar date
```

```
from datetime import date
d = date(2020, 1, 1) #2020-01-01
d.year
d.month
d.day
day = date.today()
```

Time Objects to Represent Time

```
limited to time
```

```
from datetime import time
t = time(10, 30, 50, 200555)
# time(hour, minute, second, microseconds)
# OR
t = time(hour=10, minute=30, second=50)
```

Different Between Two Times by timedelta

```
today = date(..)
new_year = date(..)
left = new_year - today

#OR

from datetime import timedelta
t1 = timedelta(weeks=.., days=.., hours=.., minutes=..,seconds=..)
t2 = ..
t3 = t2 - t1
```